

CODING TECHNICAL ASSESSMENT

Position: Full Stack Java Developer

Level: Mid-level

Technologies: Java, HTML, CSS, JavaScript

INSTRUCTIONS:

- This is a coding-focused technical assessment
- Complete all coding exercises and projects
- Submit your code files with clear documentation
- Time limit: 2-3 hours
- Reply to this email with your completed solutions

CODING TECHNICAL ASSESSMENT

Software Developer Position - Intermediate Level

****Technologies:**** JavaScript, Java

****Time Limit:**** 2-3 hours

****Instructions:**** Complete all challenges and submit working, runnable code.

Coding Exercises

Exercise 1: Algorithm Challenge

Implement a function to find the two numbers in an array that add up to a target sum.

****Requirements:****

- Function: `twoSum(nums, target)`
- Return indices of the two numbers
- Assume exactly one solution exists
- Cannot use the same element twice

****Example:****

```

Input: nums = [2, 7, 11, 15], target = 9

Output: [0, 1] (because  $\text{nums}[0] + \text{nums}[1] = 2 + 7 = 9$ )

...

### ### Exercise 2: Data Structure Implementation

Create a class that implements a queue using two stacks.

**\*\*Requirements:\*\***

- Class: `QueueWithStacks`
- Methods: `enqueue(val)`, `dequeue()`, `peek()`, `isEmpty()`
- All operations should be efficient

### ### Exercise 3: API Development

Build a REST API endpoint for managing a simple blog.

**\*\*Requirements:\*\***

- GET /api/posts - List all posts
- POST /api/posts - Create new post
- PUT /api/posts/:id - Update post
- DELETE /api/posts/:id - Delete post
- Include proper error handling and validation

**\*\*Post Schema:\*\***

```json

```
{
  "id": "string",
  "title": "string",
  "content": "string",
  "author": "string",
  "createdAt": "datetime",
  "updatedAt": "datetime"
}
```

...

Exercise 4: Database Design

Design a database schema for an e-commerce system.

****Requirements:****

- Users table (customers and admins)
- Products table
- Orders table
- Order items table
- Categories table
- Include proper relationships and constraints

****Write SQL queries for:****

1. Find top 10 best-selling products
2. Get user order history
3. Calculate monthly revenue

Practical Project

Mini Application: Task Manager

Build a full-stack task management application.

****Frontend Requirements:****

- Task list with add/edit/delete functionality
- Task status (pending, in-progress, completed)
- Due date management
- Search and filter capabilities
- Responsive design

****Backend Requirements:****

- REST API for task CRUD operations
- Data validation
- Error handling
- Simple authentication (optional)

****Technologies to use:**** JavaScript, Java

Code Review Task

Review this code and provide improvements:

```
``javascript
function getUserData(userId) {
  fetch('/api/users/' + userId)
    .then(response => response.json())
    .then(data => {
      document.getElementById('username').innerHTML = data.name;
      document.getElementById('email').innerHTML = data.email;
    })
    .catch(error => {
      console.log('Error: ' + error);
    });
}
...

```

****Tasks:****

1. Identify issues and potential problems
2. Provide an improved version
3. Explain your improvements

Submission Guidelines

1. Organize code in separate files/folders
2. Include a README with setup instructions
3. Ensure all code is runnable and tested
4. Provide clear documentation
5. Submit via email reply

****Good luck with your assessment!****

Generated on: 15/09/2025 14:30:39